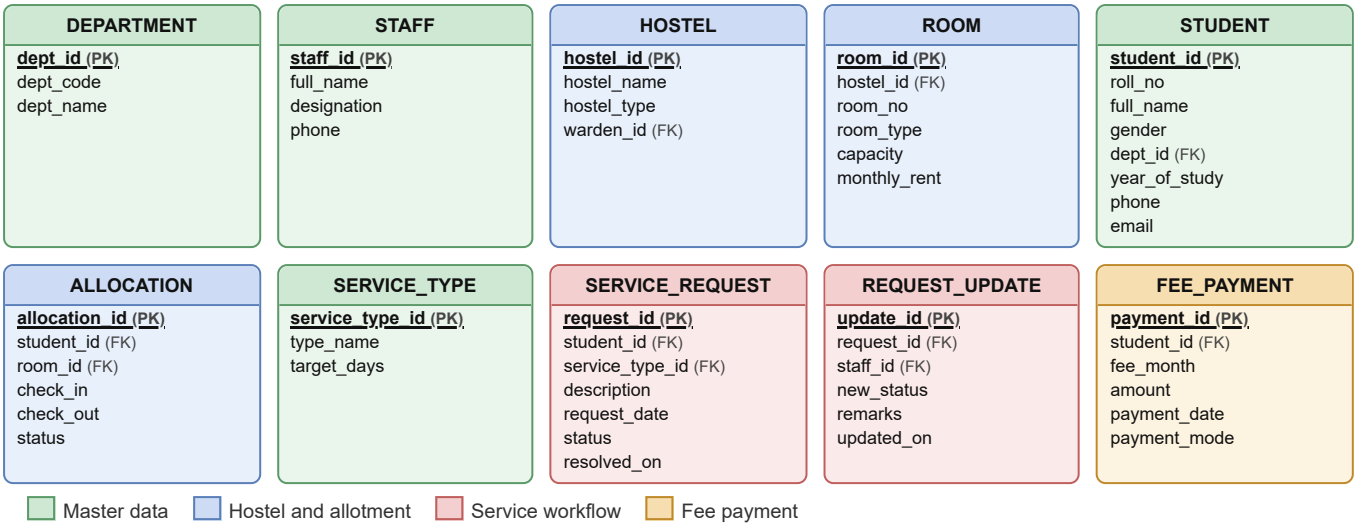
# Relational Schema

Hostel Accommodation and Student Services Management System. Ten tables, all in Third Normal Form. PK = primary key, FK = foreign key, UQ = unique.




**Primary Keys (PK)**

- DEPARTMENT.dept_id
- STAFF.staff_id
- HOSTEL.hostel_id
- ROOM.room_id
- STUDENT.student_id
- ALLOCATION.allocation_id
- SERVICE_TYPE.service_type_id
- SERVICE_REQUEST.request_id
- REQUEST_UPDATE.update_id
- FEE_PAYMENT.payment_id

**Foreign Keys (FK)**

- HOSTEL.warden_id → STAFF.staff_id
- ROOM.hostel_id → HOSTEL.hostel_id
- STUDENT.dept_id → DEPARTMENT.dept_id
- ALLOCATION.student_id → STUDENT.student_id
- ALLOCATION.room_id → ROOM.room_id
- SERVICE_REQUEST.student_id → STUDENT.student_id
- SERVICE_REQUEST.service_type_id → SERVICE_TYPE.service_type_id
- REQUEST_UPDATE.request_id → SERVICE_REQUEST.request_id
- REQUEST_UPDATE.staff_id → STAFF.staff_id
- FEE_PAYMENT.student_id → STUDENT.student_id

**Unique Keys (UQ)**

- DEPARTMENT.dept_code
- DEPARTMENT.dept_name
- STAFF.phone
- HOSTEL.hostel_name
- STUDENT.roll_no
- STUDENT.email
- SERVICE_TYPE.type_name
- ROOM(hostel_id, room_no)
- REQUEST_UPDATE(request_id, new_status)
- FEE_PAYMENT(student_id, fee_month)

## Plain text

```
DEPARTMENT(
    dept_id PK,
    dept_code,
    dept_name
)

STAFF(
    staff_id PK,
    full_name,
    designation,
    phone
)

HOSTEL(
    hostel_id PK,
    hostel_name,
    hostel_type,
    warden_id FK → STAFF.staff_id
)

ROOM(
    room_id PK,
    hostel_id FK → HOSTEL.hostel_id,
    room_no,
    room_type,
    capacity,
    monthly_rent
)

STUDENT(
    student_id PK,
    roll_no,
    full_name,
    gender,
    dept_id FK → DEPARTMENT.dept_id,
    year_of_study,
    phone,
    email
)

ALLOCATION(
    allocation_id PK,
    student_id FK → STUDENT.student_id,
    room_id FK → ROOM.room_id,
    check_in,
    check_out,
    status
)
```

SERVICE_TYPE(
  service_type_id PK,
  type_name,
  target_days
)

SERVICE_REQUEST(
  request_id PK,
  student_id FK → STUDENT.student_id,
  service_type_id FK → SERVICE_TYPE.service_type_id,
  description,
  request_date,
  status,
  resolved_on
)

REQUEST_UPDATE(
  update_id PK,
  request_id FK → SERVICE_REQUEST.request_id,
  staff_id FK → STAFF.staff_id,
  new_status,
  remarks,
  updated_on
)

FEE_PAYMENT(
  payment_id PK,
  student_id FK → STUDENT.student_id,
  fee_month,
  amount,
  payment_date,
  payment_mode
)